

Topics in CS: Problem Set 8

Due date: January 4, 2026.

Question 1. (30 points)

1. Consider the one-time pad for messages of $\ell = 3$ bits. Note that when using a key k that is all zeros, then the ciphertext is the message itself, i.e., $c = m \oplus k = m$.

There is a suggestion to make the scheme more secure by not allowing the all-zero key.

Prove that the modified scheme is no longer perfectly secret.

2. You are eavesdropping to a communication channel in which a secret message is transmitted in an encrypted form. It is known that the encryption scheme is textbook RSA with public key $N = 91$ and $e = 29$. The transmitted ciphertext is 7. What is the encrypted message?

Please write down the steps of your calculations; you can use a basic calculator.

Question 2. (30 points) Standard key-length for RSA-based schemes nowadays is $N = 2048$, where $N = pq$, which means that p and q are 1024-bit primes. In this question we will prove that instead of decrypting as $c^d \bmod N$ it is possible to work modulo the 1024-bit moduli p and q ; in practice, this gives a significant efficiency improvement. The main tool we will use is a simplified version of the *Chinese Remainder Theorem* (CRT).

1. Consider $N = 15 = 3 \cdot 5$.

- (a) Write a table with three columns: in the first list all elements of $\mathbb{Z}_{15}$, in the second the elements modulo 3, and in the third the elements modulo 5.
- (b) Add the rows corresponding to 4 and 2. How does the result relate to the row for 6? Multiply the rows corresponding to 4 and 2. How does the result relate to the row for 8? What general property of the mapping $f(z) = (z \bmod p, z \bmod q)$ does this suggest?
- (c) Denote by (x, y) the row where the second element is x and the third is y . In what sense is $(2, 3) = 2(1, 0) + 3(0, 1)$ in $\mathbb{Z}_{15}$?
- (d) Run the extended Euclidean algorithm on 5 and 3. What is the connection between the coefficients found and the first element in the rows corresponding to $(1, 0)$ and $(0, 1)$?

2. Let p and q be distinct primes, denote $N = pq$, and consider the function $f : \mathbb{Z}_N \rightarrow \mathbb{Z}_p \times \mathbb{Z}_q$ defined by $f(z) = (z \bmod p, z \bmod q)$. Prove that $f : \mathbb{Z}_N \rightarrow \mathbb{Z}_p \times \mathbb{Z}_q$ is a homomorphism.
3. Prove that f is invertible and find its inverse $f^{-1} : \mathbb{Z}_p \times \mathbb{Z}_q \rightarrow \mathbb{Z}_N$.

Hint: for $(x, y) \in \mathbb{Z}_p \times \mathbb{Z}_q$ denote $(x, y) = x(1, 0) + y(0, 1)$; which elements in $\mathbb{Z}_N$ correspond to $(1, 0)$ and $(0, 1)$?

4. So far we have concluded that the additive groups are isomorphic $\mathbb{Z}_N \cong \mathbb{Z}_p \times \mathbb{Z}_q$. Prove that the multiplicative groups are also isomorphic, i.e., that $\mathbb{Z}_N^* \cong \mathbb{Z}_p^* \times \mathbb{Z}_q^*$.
5. Describe the modified decryption algorithm for textbook RSA in which exponentiations modulo N ($c^d \bmod N$) are replaced with exponentiations modulo p and q ($c^d \bmod p$ and $c^d \bmod q$).

Question 3. (10 points) Use the CRT to find $x \in \mathbb{Z}_{77}$ such that $x \equiv 5 \pmod{7}$ and $x \equiv 8 \pmod{11}$.

Question 4. (30 points) Implement the Textbook RSA algorithms: the key-generation algorithm, the encryption algorithm, and the modified decryption algorithm using the CRT. Use the SamplePrime and Extended Euclidean algorithms you implemented in prior assignments.

Run the key-generation algorithm for $n = 512$ and $e = 3$. Sample a random number $m \leftarrow \mathbb{Z}_N$, encrypt it as $c = \text{Enc}_{pk}(m)$ and verify that indeed $m = \text{Dec}_{sk}(c)$. Write the values that you receive.

Good luck!